Article

Creating specialized classes for fire risk using machine learning methods

Constantina Kopitsa[1], Ioannis G. Tsoulos[2], Vasileios Charilogis[3], Glykeria Kyrou[4*]

[1] Department of Informatics and Telecommunications. University of Ioannina,
[2] Department of Informatics and Telecommunications. University of Ioannina,
[3] Department of Informatics and Telecommunications. University of Ioannina,
[4] Department of Informatics and Telecommunications. University of Ioannina
* Correspondence: g.kyrou@uoi.gr

Abstract

This study explores various machine learning techniques for assessing forest fire risk. The data used in this study have been selected from three different areas in Greece over the past 25 years (2000 - 2024). The first area of interest is Attica, the second area is Eastern Macedonia and Thrace, and the third area is Crete. For each of the aforementioned areas, a separate data file was created, an analysis was carried out, and specialized definitions - classes were created for the forest fire risk occurrence. This study utilized a range of traditional machine learning techniques alongside novel methods based on Grammatical Evolution, including automated feature construction from existing attributes and the automatic generation of classification rules. Across all three datasets, GENCLASS demonstrated the most consistent overall performance among the tested techniques, closely followed by the feature construction method FC2 (MLP), while the remaining traditional approaches (MLP, RBF, and Random Forest) showed comparatively lower and more variable performance. These findings validate the climatological attributes and region specific definitions developed for the classification of fire danger, confirming that region specific thresholds provide a more reliable framework for wildfire risk assessment than the generic 30/30/30 rule.

Keywords: Fire risk; Machine learning; neural networks

1. Introduction

This study presents a new rule for forest fire risk, which specifies and adapts the climatic and forest fire data with a new framework instead of the well known 30/30/30. In short the rule 30/30/30 means that there is a risk of fire when the following conditions prevail: temperatures >30 , wind >30, and humidity <30. However, at this point we will briefly give a sample of analyses and experiments, with new definitions, which are specific to each region. For example, the dangerous mean temperature point for Attica is 23.6 °C or 74.5 °F, for Eastern Macedonia and Thrace is 20 °C or 68.1 °F, and for Crete is 20.17 °C or 68.3 °F. In other words, wildfires do not necessarily follow our predefined rules, however, our newly developed, region - with specific rules - adapted to the unique characteristics of each area, can contribute to more effective fire prevention and suppression efforts. In this context, it is important to highlight several real world case studies, that demonstrate the necessity of establishing new, region specific thresholds for wildfire ignition risk.

- A first example is the wildfire that broke out during the winter season, specifically in January 2025, in California, lasting 24 days [1]. Moreover, the study by Bouvier et al.

Received:
Revised:
Accepted:
Published:

Copyright: © 2026 by the authors. Submitted to *Journal Not Specified* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license.

https://doi.org/10.3390/1010000

(2025) indicates that an improved wildfire prevention model is required for periods traditionally considered as non hazardous for fire occurrence [2]. It is worth noting that, based on the meteorological data we retrieved from Visual Crossing for the region of California for the period 01/01/2025 - 31/01/2025, for this event, the mean temperature was approximately 5 °C (42 °F), with maximum temperatures reaching only 15 °C (60 °F)[3].

- A second example is the wildfire that broke out during the early spring season, specifically in March 2022, in South Korea [4]. Notably, by examining the temperature records for that specific period in the region of South Korea, the nighttime minimum temperatures dropped to -4 °C (24.8 °F), while daytime maximum temperatures ranged between 12 °C (53.6 °F) and 14 °C (57.2 °F) [5].
- A third example is the wildfire that occurred in May 2025 near the city of Chita in Russia [6]. During that period, temperatures were approximately 17.2 °C (63 °F), with the lowest recorded temperature reaching 4.4 °C (40 °F) [7].
- A fourth example is the winter wildfire that occurred in Ofunato, Japan, in February 2025 [8]. During that period, temperatures ranged from −1.6 °C (29°F) to a maximum of 3.8 °C (39 °F) [9].

In summary, the aforementioned incidents represent only a subset of the wildfire events identified through the literature review, all of which fall outside the scope of the traditional 30/30/30 rule. This highlights the need to develop and examine new, region specific and context adapted metrics for assessing wildfire occurrence risk, based on meteorological variables and operational firefighting data.

In our search for studies related to our own work, we initially focused on identifying research that examined the application of the 30/30/30 rule and its reported outcomes, using the keywords: *30/30/30 rule fire risk.* The review of the retrieved studies indicated that only one verification oriented investigation of the rule has been conducted, and its findings were not satisfactory. Specifically, the study by Cuevas and Rodriguez (2021) highlights that the 30/30/30 rule does not account for the majority of large scale wildfires, explaining only about 10% of such events. In contrast, it appears more relevant for medium sized wildfires, accounting for approximately 66% of these cases. Moreover, temperature and relative humidity emerge as the most critical variables, as over 80% of the analyzed wildfires met these two conditions [10]. We will proceed by presenting studies and research works that examine the impact of meteorological variables and modeling approaches on forest fires, comparable to our own, and, where applicable, we will conclude by comparing their reported performance metrics with the findings of the present study.

The study by Cortez and Morais (2007) employed meteorological variables such as: temperature, relative humidity, rainfall, and wind, using an SVM model capable of predicting the burned area of small fires. Their approach achieved accurate predictions in 46% of the cases when an error of ≤ 1 ha was allowed, and in 61% of the cases when the admissible error increased to ≤ 2 ha [11]. The publication by Yunhao et al. (2007) focuses on comparing two wildfire risk indices: the Fire Potential Index (FPI) and the Integrated Fire Susceptibility Index (IFSI). Their experimental results, based on the Area Under the Curve (AUC) of the REC curves, showed that IFSI performed as the superior model with an accuracy of approximately 78%, whereas FPI achieved a prediction accuracy of about 73% [12].

Next on the research by Sakr et al. (2011) employed only two meteorological variables cumulative precipitation and relative humidity, together with two artificial intelligence techniques, artificial neural networks (ANN) and support vector machines (SVM), to accurately predict the risk of fire occurrence. Their results showed that, for the SVM model, each month included at least one iteration in which the testing accuracy exceeded 97%

Table 1. Comparative overview of accuracy/performance results reported by previous studies on forest fire risk prediction, as identified in the literature review.

Authors names	Results
Diaz Cuevas, M. D. P., & Limones Rodriguez, N.[10]	10% accuracy (large fires) & 66% (medium fires)
Cortez, P., & Morais, A. D. J. R[11]	46% error (of $\leq$ 1ha) & 61% ($\leq$ 2 ha)
Yunhao, C., Jing, L., & Guangxiong, P.[12]	IFSI accuracy (78%) & FPI (73%)
Sakr, G. E., Elhajj, I. H., & Mitri, G.[13]	SVM accuracy (97%)
Agarwal, P., Tang, J., Narayanan, A. N. L., & Zhuang, J[14]	Gradient Boosting Tree accuracy (93.3%) & test set (97%)
Masinda, M. M., Li, F., Qi, L., Sun, L., & Hu, T[15]	FWI Wilcoxon test accuracy (72.7%)
Karali, A., Varotsos, K. V., Giannakopoulos, C., Nastos, P. P., & Hatzaki, M[16].	FWI & ISI accuracy (60%)
Parvar, Z., Saeidi, S., & Mirkarimi, S.[17]	AUC value of 0.83

[13]. Analogous result were reported in the study by Agarwal et al. (2020) who applied the Gradient Boosting Tree algorithm using data from the National Fire Incident Reporting System (NFIRS) together with meteorological information from the National Oceanic and Atmospheric Administration (NOAA) to predict economic losses caused by fire incidents. Their results demonstrated strong predictive performance, achieving 93.3% accuracy on the full dataset and 97% accuracy on the test set [14]. The subsequent field study by Masinda et al. (2022) examined the daily assessment of wildfire danger in Northeastern China using the Canadian Forest Fire Weather Index (FWI). Their results indicated that the Canadian FWI performed well, with the Wilcoxon paired sample test showing satisfactory agreement in 72.7% of the cases [15]. The same modeling framework, combining the Canadian Forest Fire Weather Index (FWI) with the Initial Spread Index (ISI), was examined by Karali et al. (2023) in their assessment of hazardous wildfire prone conditions. Their findings indicated that lead time plays a decisive role in forecast performance, with the probability of correct prediction reaching approximately 60% [16]. In their study, Parvar et al. (2024) developed a wildfire risk map by integrating meteorological variables such as: Land Surface Temperature (LST), precipitation, and historical fire occurrence data. The performance of their model was statistically evaluated using the ROC curve, achieving an AUC value of 0.83%, indicating strong predictive capability [17].

At this point, we will present, in a table 1, the results of the aforementioned studies in order to enable a direct comparison among them Table 1. In Section4 a concise comparison will also be provided between the findings of each related study and those of our own research.

The remaining of this paper has the following structure: the Section 2 presents the used dataset and the new threshold values for wildfire occurrence, the Section 3 illustrates the experimental results finally the Section 4 It provides a concise presentation of the results, accompanied by a comparative discussion with the previously cited studies in the table 1.

2. Materials and Methods

In this section, a detailed presentation of the datasets used as well as the machine learning techniques used in the experiments performed will be provided.

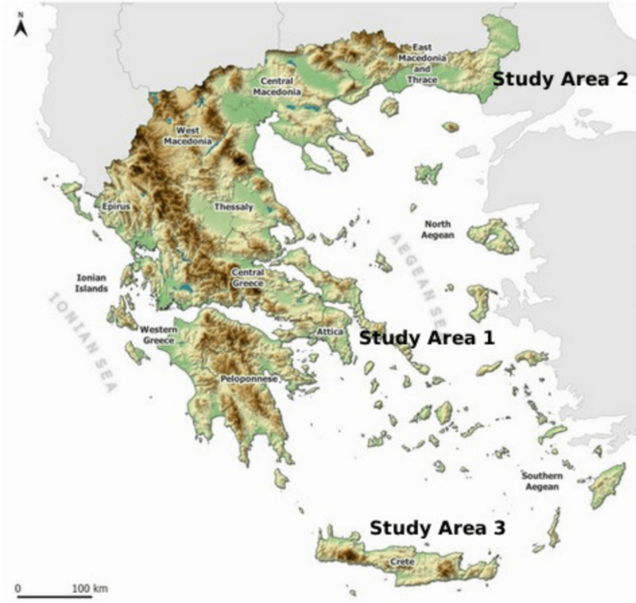

Figure 1. Map of Greece highlighting the three geographic study areas (Study Area 1: Attica, Study Area 2: Eastern Macedonia & Thrace, Study Area 3: Crete).

2.1. The dataset employed

For the purposes of our study, we combined two distinct data sources. On the one hand, to analyze the climatological conditions, we employed Historical Weather Data from Visual Crossing https://www.visualcrossing.com/[3] (last accessed on 02 February 2026), which compiles information from a range of authoritative sources, including national meteorological agencies, global climate datasets, and satellite observations. Access to the historical dataset required completing the registration process, after which we selected the data-access interface, specified the region of interest, and defined the temporal range. Data retrieval was then carried out under the free access tier, which allows up to 1,000 records per day. The procedure is generally straightforward, with the only notable constraint being the daily limit of 1,000 data points.

On the other hand, for the analysis of forest / wild fires, we relied on the open data provided by the Hellenic Fire Service https://www.fireservice.gr/el_GR/synola-dedomenon (last accessed on 02 February 2026). The fire data are collected and updated each year by the individual fire stations responsible for reporting incidents.

2.2. The dataset description

For this study, we obtained historical meteorological data for the period 2000 – 2024, providing a roughly 25 year record to support our fire duration according to climate patterns analysis. We additionally selected regions across different latitudes and longitudes, in Greece, to achieve a more comprehensive representation of the study areas. The selected regions of interest are: Attica, Eastern Macedonia and Thrace, and Crete 1.

The climatic and fire dataset for all regions covers the period from January 1, 2000, to December 31, 2024. Specifically, Attica provides 9.131 climatic dataset and 2.736 fire records, Eastern Macedonia and Thrace provides 9.131 climatological records and 10.358 fire records, and finally Crete provides 9.121 climatological records and 6.036 fire record. The variation in the number of climatological records is attributable to gaps (missing values) in the data retrieved from Visual Crossing, which did not influence our results. Before proceeding with the merging of the two data files, the experiments focused exclusively on forest fire incidents in order to maintain homogeneity in our dataset with respect to fire behavior characteristics. In other words, we chose to proceed with the following categories : (9) Burned forests, (10) Burned forest area, and (11) Burned grove (small forest) Table 3. Following this refinement, Attica provided 1.859 forest fire records, Eastern Macedonia and Thrace 2.652, and Crete 2.787.

Afterwards, the climatological dataset merged with the forest fire dataset, retaining only the climate records corresponding to the fire ignition date. The merging process was performed combining the climatological data Table 2 with the fire incident data Table 3 based on the date field.

Table 2. Primary table of raw climatological variables (33 attributes) retrieved from the Visual Crossing platform for the period 2000–2024.

Number	Data Description	Values
1	Region	Area
2	Date	Year/month/day
3	Tempmax (Fahrenheit)	°F
4	Tempmin (Fahrenheit)	°F
5	Temp (Fahrenheit) daily mean average	°F
6	Feelslikemax (Fahrenheit)	°F
7	Feelslikemin (Fahrenheit)	°F
8	Feelslike (Fahrenheit)	°F
9	Dew	°F
10	Humidity	%
11	Precipitation	in
12	Precipprob	%
13	Precipcover	%
14	Preciptype	Type(s)
15	Snow	in
16	Snowdepth	in
17	Windgust (miles)	mph
18	Windspeed (miles)	mph
19	Winddir	(°)
20	Sealevel	mb
21	Cloudcover	%
22	Visibility	mi
23	Solarradiation	(W/m^2)
24	Solarenergy	(MJ/m^2)
25	Uvindex	(0-10)
26	Severisk	(0-100)
27	Sunrise	time
28	Sunset	time
29	Moonphase	(0-1)
30	Conditions	general observations
31	Descriptions	general observations day/night
32	Icon	summarized observations
33	Stations	Code for stations

https://doi.org/10.3390/1010000

Table 3. Primary table of raw operational forest fire incident data (16 attributes) as recorded by the Hellenic Fire Service.

Number	Data Description
1	Fire department
2	Forest department
3	Location
4	Region
5	Date of fire ignition
6	Time of fire ignition
7	Extinguish date
8	Extinguish time
9	Burned forests
10	Burned forests area
11	Burned grove (small forest)
12	Burned grassland
13	Burned swamps
14	Burned agricultural
15	Burned crops
16	Burned garbage dumps

Accordingly, we created three new datasets, one for each study region. Each region was examined and analyzed independently, resulting in a distinct fire risk classification for every area. Consequently, the variables processed in order to construct the class representing danger forest fire ignition, for all three datasets were the following: mean daily temperature, daily humidity values, precipitation occurrence, wind gusts, wind speed, wind direction, time of fire ignition, month of fire ignition, and fire duration, with the final output expressed as the Final_DangerClass_combination Table 4. In the end, in order to ensure impartiality in our experiments, we removed the FireDuration feature, since the model would otherwise be able to see the answer, as the accuracy for all datasets reached 100%.

Table 4. Final derived variables/classes obtained after processing the raw data, used to construct the target variable Final_DangerClass_combination.

Number	Data
1	TempSpecialClass (daily mean average)
2	HumiditySpecialClass
3	PrecipitationBasicClass
4	WindgustSpecialClass
5	WindspeedSpecialClass
6	WinddirSpecialClass
7	TimeClass
8	Monthclass
9	Fireduration (Removed)
10	Final_DangerClass_combination

2.2.1. Attica area and data description

The regional unit of Attica covers an area of 3,808.1 km², with Athens as its capital. Between 2001 and 2025, Attica experienced a loss of 22 kha (220 km²) of tree cover attributable to wildfires, along with an additional 3.3 kha (33 km²) lost due to other disturbance factors. The most severe year within this period was 2021, during which fires accounted for 8.5 kha (85 km²) of tree cover loss, representing 99% of the total loss recorded that year, according to the global forest watch [18].

A brief description of the Attica dataset follows. The final dataset consisted of 1.859 records and 8 attributes, with the last attribute representing the target class, namely: Final_DangerClass_combination. The classes followed a binary scheme, where 0 denoted a non hazardous value, and 1 denoted as a hazardous value, in the following Table 5.

Table 5. Definition of the threshold values (temperature, humidity, wind, wind direction, ignition time, month) separating the non-hazardous (Class 0) from the hazardous (Class 1) fire danger class for the Attica region, with the corresponding number of records per class.

Features	**Class 0 (non hazard)**	**Class 1 (hazard)**
TempSpecialClass (daily mean average)	<= (73.6°F)	73.7°F = > (or 23.6°C)
HumiditySpecialClass	49.4% = >	< = 49.3%
PrecipitationBasicClass	rain	no rain
WindgustSpecialClass	< = 10miles	10.01miles =>
WindspeedSpecialClass	< = 16miles	16.1miles = >
WinddirSpecialClass	(S, SE, E, W)	(SW, N, NE, NW)
TimeClass	< = 12:30	12:31 =>
Monthclass	(1,2,3,4,5,9,10,11,12)	(6,7,8*)
Final_DangerClass_combination	844 data	1.015 data

In summary, the findings of the study identified new threshold values for wildfire occurrence in the region of Attica, as follows: temperature $\geq$ 73.7 °F, humidity $\leq$49.3%, 0% precipitation, wind gust speed $\geq$10.01 mph, wind speed $\geq$16.1 mph, wind direction in the sectors SW, N, NE, and NW, ignition time from 12:31 on wards, and hazardous months corresponding to June, July, and August*.

2.2.2. Eastern Macedonia and Thrace area and data description

The regional unit of Eastern Macedonia and Thrace, covers an area of 14.158 km^2, with Komotini as its capital. Between 2001 to 2025, Eastern Macedonia and Thrace experienced a loss of 40 kha (400 km^2) of tree cover attributable to wildfires, along with an additional 11 kha (110 km^2) lost due to other disturbance factors. The most severe year within this period was 2023, during which fires accounted for 31 kha (310 km^2) of tree cover loss, representing 94% of the total loss recorded that year, according to the global forest watch[18].

A brief description of the Eastern Macedonia and Thrace dataset follows. The final dataset consisted of 2.653 records and 8 attributes, with the last attribute representing the target class, namely: Final_DangerClass_combination. The classes follow the same binary scheme as previously. A concise summary table of these variables is presented in Table 6.

Table 6. Definition of the threshold values (temperature, humidity, wind, wind direction, ignition time, month) separating the non-hazardous (Class 0) from the hazardous (Class 1) fire danger class for the Eastern Macedonia & Thrace region, with the corresponding number of records per class.

Features	**Class 0 (non hazard)**	**Class 1 (hazard)**
TempSpecialClass (daily mean average)	< = 68°F	68.1°F = >(or 20°C)
HumiditySpecialClass	64.1% = >	< = 64%
PrecipitationBasicClass	rain	no rain
WindgustSpecialClass	< = 7miles	7.1miles =>
WindspeedSpecialClass	< = 6miles	6.1miles =>
WinddirSpecialClass	(W,NW,N)	(E,SE,SW,S,NE)
TimeClass	< = 11:30	11:31 =>
Monthclass	(1,2,3,4,9,11,12)	(5,6,7,8,10*)
Final_DangerClass_combination	1.093	1.559

In summary, the findings of the study identified new threshold values for wildfire occurrence in the region of Eastern Macedonia and Thrace, as follows: temperature $\geq$ 68.1 °F, humidity $\leq$ 64.1%, 0% precipitation, wind gust speed $\geq$ 7.1 mph, wind speed $\geq$ 6.1 mph, wind direction in the sectors E, S, SW, SE, and NE, ignition time from 11:31 on wards, and hazardous months corresponding to May, June, July, August and October.

2.2.3. Crete area and data description

The regional unit of Crete covers an area of 8.450 km², with Herakleio as its capital. Between 2001 to 2025 Crete experienced a loss of 1.3 kha (13 km²) of tree cover attributable to wildfires, along with an additional 3.2 kha (32 km²) lost due to other disturbance factors. The most severe year within this period was 2008 during which fires accounted for 210 ha (2.1 km²) of tree cover loss, representing 44%, of the total loss recorded that year, according to the global forest watch [18].

A brief description of the Crete datasets follows in Table 7.

Table 7. Definition of the threshold values (temperature, humidity, wind, wind direction, ignition time, month) separating the non-hazardous (Class 0) from the hazardous (Class 1) fire danger class for the Crete region, with the corresponding number of records per class.

Features	Class 0 (non hazard)	Class 1 (hazard)
TempSpecialClass (daily mean average)	< = 70°F	70.1°F = >(or 20.17°C)
HumiditySpecialClass	66.1% = >	< = 66%
PrecipitationBasicClass	rain	no rain
WindgustSpecialClass	< = 9miles	9.1miles =>
WindspeedSpecialClass	< = 16miles	16.1miles =>
WinddirSpecialClass	(S, SW, SE, E, W)	(N, NE, NW)
TimeClass	< = 12:30	12:31 =>
Monthclass	(1,2,3,4,5,9,10,11,12)	(6,7,8*)
Final_DangerClass_combination	1.383	1.404

In summary, the findings of the study identified new threshold values for wildfire occurrence in the region of Crete, as follows: temperature $\geq$ 70.1 °F, humidity $\leq$66%, 0% precipitation, wind gust speed $\geq$9.1 mph, wind speed$\geq$ 16.1 mph, wind direction in the sectors N, NE, and NW, ignition time from 12:31 on wards, and hazardous months corresponding to June, July, and August .

3. Results

For the purposes of our experiments, we employed the available optimization package Optimus[19], which can be downloaded fromhttps://github.com/itsoulos/GlobalOptimus (accessed on 20 June 2026). For the feature construction process used in the FC2(RF) and FC2(MLP) algorithms, we utilized the freely available programming tool QFc[20], accessible at https://github.com/itsoulos/QFc (accessed on 20 June 2026). In addition, the GENCLASS [21] system was employed in our experiments. GENCLASS is a free software framework entirely written in ANSI C++, available by https://github.com/itsoulos/GenClass/(accessed on 20 June 2026) designed to construct classification programs in a C like programming language for the purpose of classifying input data. It operates as an evolutionary rule induction method, generating interpretable decision rules through genetic search mechanisms.

Each experiment was executed 30 times, using different seeds for the random number generator in every run. In addition, the ten-fold cross-validation procedure was applied to ensure robust validation of the experimental results. The proposed feature construction method was applied independently to each fold, and the average classification errors on the corresponding test sets were computed. For the experimental tables the following notation was used:

1. The column DATASET denotes the dataset corresponding to a regions where the machine learning methods were applied. Attica contains 1.859 records, of which 844 correspond to class 0 and 1.015 to class 1. Eastern Macedonia & Thrace contains 2.653 records, of which 1.093 correspond to class 0 and 1.559 to class 1. Crete contains 2.789 records, of which 1.383 correspond to class 0 and 1.404 to class 1.
2. The column MLP(BFGS) represents the application of an artificial neural network [23,24] with 10 processing nodes trained with the BFGS optimization method [25].
3. The column RBF stands for the usage of a Radial Basis Function (RBF) network [26,27] with 10 processing nodes.
4. The column RF denotes the application of the Random Forest model [28].
5. The column FC2(RF) represents the construction of two artificial features using the Grammatical Evolution feature construction technique [29]. The Radial Basis Function (RBF) machine learning model was employed for feature construction, while the generated features were evaluated using Random Forests.
6. The column FC2(MLP) stands for the construction of two artificial features using the same feature construction technique. The evaluation of the constructed features was performed using a neural network trained with the BFGS method.
7. The column GENCLASS represents the incorporation of the GENCLASS method to produce classification rules.
8. The row AVERAGE represents the average classification error for each dataset and method.

3.1. Classification error

The results demonstrated in Table 8 that the model GENCLASS achieved the lower error accuracy in predicting the final danger class, indicating that the newly developed fire danger rule is fully identifiable and predictable based on the combined climatological and operational fire incident data.

Table 8. Experimental results using the machine learning methods on various regions. The numbers in cells represent average classification error.

DATASET	MLP(BFGS)	RBF	RF	FC2(RF)	FC2(MLP)	GENCLASS
Attica	13.67%	14.42%	13.93%	13.01%	12.82%	12.43%
Eastern Macedonia & Thrace	12.51%	15.34%	12.21%	11.93%	11.87%	11.47%
Crete	16.39%	17.82%	15.96%	16.15%	15.56%	14.86%
AVERAGE	**14.19%**	**15.86%**	**14.03%**	**13.70%**	**13.42%**	**12.92%**

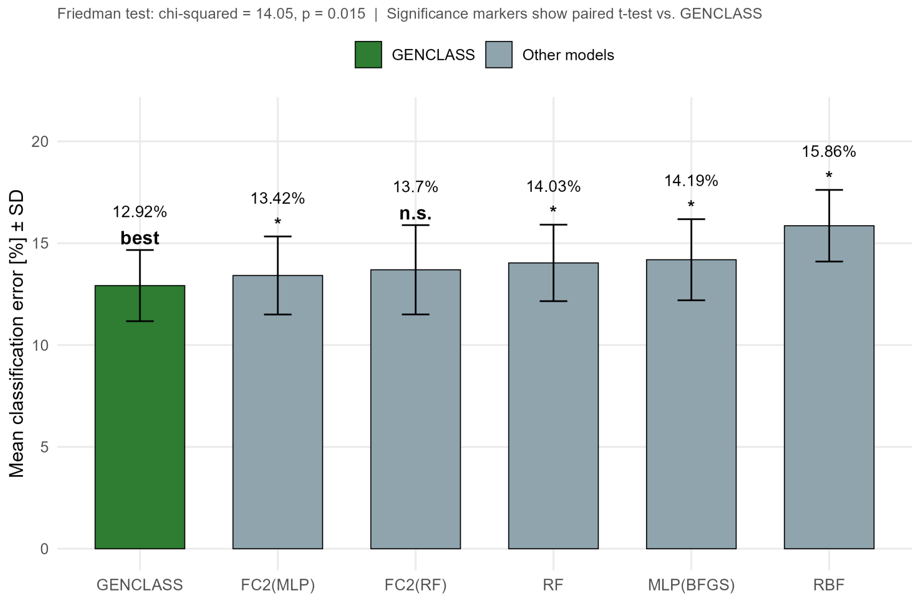

Figure 2. Average classification error by , across the three study regions

Figure 2 presents, in bar-chart form, the mean classification error rate (%) achieved by each of the six evaluated machine learning models, averaged across the three study regions (Attica, Eastern Macedonia & Thrace, and Crete), with error bars representing one standard deviation. GENCLASS is visually distinguished from the remaining five models and serves as the reference model against which pairwise comparisons are drawn. It attained the lowest mean classification error (12.92%), followed in ascending order by FC2(MLP) (13.42%), FC2(RF) (13.70%), RF (14.03%), MLP(BFGS) (14.19%), and RBF (15.86%), the latter exhibiting both the highest mean error and the greatest variability across regions. The figure further reports the result of a Friedman test (χ^2 = 14.05, df = 5, p = 0.015), a non-parametric procedure appropriate for comparing multiple related samples, which indicates that the observed differences in performance among the six models are unlikely to have arisen by chance. Significance markers displayed above each bar correspond to paired t-test comparisons against GENCLASS: all models differ significantly from GENCLASS ($p < 0.05$) with the exception of FC2(RF), which is marked as not significant (n.s.), suggesting that its performance is statistically comparable to that of GENCLASS despite its marginally higher mean error.

Table 9. Statistical comparison of classification error, average ranking, and significance tests (Wilcoxon and paired t-test) for the six machine learning models relative to GENCLASS as the reference model.

Model	Mean Error	SD Error	Min Error	Max Error	Avg Rank	Mean Diff vs GENCLASS	Wilcoxon p	Paired p	Significance
GENCLASS	12.92	1.75	11.47	14.86	1	0			Reference
FC2(MLP)	13.42	1.92	11.87	15.56	2	-0.5	0.1814	0.0395	*
FC2(RF)	13.7	2.19	11.93	16.15	3.33	-0.78	0.1814	0.0955	n.s.
RF	14.03	1.88	12.21	15.96	4	-1.11	0.1814	0.0367	*
MLP(BFGS)	14.19	1.99	12.51	16.39	4.67	-1.27	0.1814	0.0123	*
RBF	15.86	1.76	14.42	17.82	6	-2.94	0.1814	0.0324	*

Friedman test: chi-squared = 14.048, df = 5, p-value = 0.0153
Nemenyi critical difference (alpha = 0.05): 4.353
*Significance vs. GENCLASS (paired t-test): *** p<0.001, ** p<0.01, * p<0.05, n.s. = not significant*
Note: n = 3 regions per technique; results are indicative given the small sample size.

Table 9 complements Figure 2 by providing a detailed statistical summary of model performance across the three regions (n = 3 per model). For each model, the table reports the mean, standard deviation, minimum, and maximum classification error, together with the average rank obtained across regions, the mean difference relative to GENCLASS, and the corresponding significance estimates derived from both the Wilcoxon signed-rank test and the paired t-test. GENCLASS consistently ranks first (average rank = 1) with the lowest mean error and a comparatively narrow standard deviation (SD = 1.75), whereas RBF ranks last (average rank = 6) with both the highest mean error and the widest range of observed values. The Wilcoxon test yields an identical p-value (0.1814) for all comparisons, which is not statistically significant; this outcome reflects the limited statistical power of the test given the very small sample size (n = 3 paired observations per comparison). In contrast, the paired t-test, being a parametric test with greater sensitivity under these conditions, identifies statistically significant differences ($p < 0.05$) between GENCLASS and all other models except FC2(RF) (p = 0.0955, n.s.), which again emerges as the closest competitor to GENCLASS. The table footnote reports the overall Friedman test statistic (χ^2 = 14.048, df = 5, p = 0.0153) and the Nemenyi critical difference (4.353) used for post-hoc multiple comparisons, while explicitly cautioning that, owing to the small sample size (three regions per technique), the results should be regarded as indicative rather than conclusive.

Taken together, Figure 2 and Table 2 converge on the conclusion that GENCLASS systematically outperforms the remaining five models, with the superiority being statistically robust, as confirmed by the paired t-test, against every model except FC2(RF), which stands out as its most competitive alternative.

3.2. Precision and recall values

Precision and recall are evaluation metrics used to assess the performance of a machine learning model. Precision expresses the proportion of predicted positive cases that are actually correct, while recall indicates the proportion of true positive cases that the model successfully identifies. Together, these metrics provide essential insight into the effectiveness of a model in classification tasks. Precision is calculated as:

$$\text{Precision} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalsePositives}} \tag{1}$$

Precision reflects the model's ability to avoid false positives, making it a key indicator of prediction reliability. In other words, the higher the value, the fewer the errors.

The Recall value confirms the model's ability to correctly identify the positive class, reflecting how many of the actual positive instances were successfully detected. As with precision and recall, the higher the value, the more successful the classification. Recall is calculated as:

$$\text{Recall} = \frac{\text{TruePositives}}{\text{TruePositives} + \text{FalseNegatives}} \tag{2}$$

The following table presents the average values of the precision and recall measures for each dataset and for every method Table 10. In this comparison as well, we observe the consistent performance of all methods. The best values are achieved by RF, FC2(RF), FC2 (MLP), and GENCLASS, while for the Eastern Macedonia & Thrace dataset, MLP (BFGS) stands out.

Table 10. Comparative summary of average precision and recall values (%) for each machine learning technique, by study region (Attica, Eastern Macedonia & Thrace, Crete).

		MLP (BFGS)	RBF	RF	FC2 (RF)	FC2 (MLP)	GENCLASS
Attica	Precision	87.98%	85.43%	88.07%	88.24%	88%	88.50%
	Recall	87.14%	85.32%	87.33%	87.32%	88.02%	88.10%
Eastern Macedonia	Precision	89.04%	83.51%	89.45%	87.92%	88.08%	89%
	Recall	88.05%	83.82%	88.37%	89.08%	89.08%	88.9%
Crete	Precision	84.46%	82.09%	85.07%	84.95%	84.46%	85.07%
	Recall	84.39%	82.09%	85.03%	84.88%	84.52%	85.50%

The precision and recall values presented in Table 10 confirm the pattern already observed for classification error, showing consistently high and stable performance across all three regions. For the Attica dataset, both metrics ranged between 85.32% and 88.50%, with GENCLASS, FC2(RF), and RF achieving the strongest results, while RBF again recorded the lowest performance, at approximately 85%. The Eastern Macedonia & Thrace dataset exhibited the highest overall performance levels among the three regions, ranging from 83.51% to 89.45%, with RF, MLP (BFGS), and GENCLASS standing out in precision, and FC2(RF) and FC2(MLP) achieving the highest recall values. In contrast, the Crete dataset showed comparatively lower performance levels, ranging from 82.09% to 85.50%, although the relative ranking of the techniques remained consistent with the other two regions, with GENCLASS and RF standing out. Across all three datasets, RBF consistently underperformed relative to the other five techniques, whereas GENCLASS, RF, and the feature construction methods FC2(RF) and FC2(MLP) maintained the most stable and competitive precision and recall values, corroborating the reliability of the region specific fire danger thresholds developed in this study.

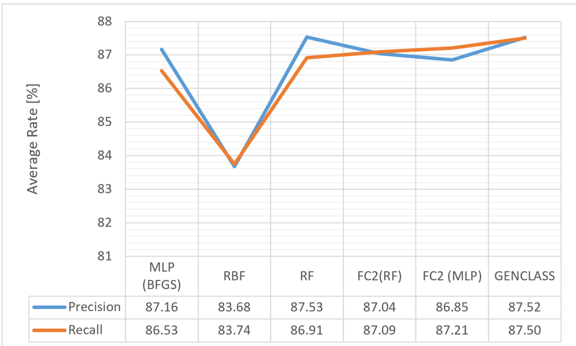

Figure 3. Line chart of the average precision and recall (%) across the three study regions, for each machine learning model.

Figure 3 summarizes these findings, showing that GENCLASS achieves the highest average recall across the three regions (87.50%) and an average precision (87.52%) essentially equivalent to that of RF, the top-performing model on this metric (87.53%), while RBF remains the weakest performing technique throughout.

4. Conclusions

This study developed and validated region specific fire danger classification schemes for three climatically distinct areas of Greece, namely Attica, Eastern Macedonia & Thrace, and Crete, using twenty five years (2000-2024) of combined climatological and operational fire incident data. Rather than relying on the generic 30/30/30 rule, specialized threshold values for temperature, humidity, wind gust, wind speed, wind direction, ignition time, and month were derived independently for each region, reflecting their distinct climatic profiles.

Six machine learning techniques were evaluated across all three datasets using ten fold cross-validation: two classical approaches (MLP trained with BFGS, and RBF networks), Random Forest (RF), two Grammatical Evolution based feature construction methods (FC2(RF) and FC2(MLP)), and GENCLASS, an evolutionary rule induction method also grounded in Grammatical Evolution. The results consistently demonstrated a clear performance hierarchy: GENCLASS achieved the lowest average classification error (12.92%) and the highest average precision and recall values across all three regions, confirming that the fire danger rule constructed in this study is fully identifiable and predictable from climatological and operational data. FC2(MLP) and FC2(RF) followed closely, ranking second and third respectively in average classification error (13.42% and 13.70%), while the classical techniques (RF, MLP (BFGS), and particularly RBF) exhibited comparatively higher and more variable error rates, with RBF consistently ranking last across all three regions in both classification error and precision/recall metrics.

This pattern indicates that techniques grounded in Grammatical Evolution, whether through automated feature construction (FC2) or direct rule induction (GENCLASS), are better suited to capturing the complex, region specific relationships between meteorological conditions and wildfire occurrence than conventional neural network or ensemble based approaches. In particular, the superiority of GENCLASS suggests that the interpretable decision rules it generates are not only competitive in predictive accuracy but also offer a transparent and operationally meaningful representation of fire danger, a property that black box models such as MLP and RBF cannot provide. These findings validate the climatological attributes and the region specific definitions developed for the classification of fire danger, confirming that tailored, evolutionarily derived thresholds provide a more reliable and robust framework for wildfire risk assessment than generic, one size fits all rules.

Building on these findings, several directions merit further investigation. First, the application of the proposed methodology to additional Greek regions with different climatic and topographic characteristics would help establish whether the observed superiority of Grammatical Evolution based techniques generalizes beyond the three areas examined here. Second, incorporating additional predictive features, such as vegetation type, fuel moisture content, terrain slope, and proximity to human activity, could further improve the discriminative power of the constructed classes, particularly for the Crete dataset, which exhibited comparatively lower performance across all techniques. Third, future work could explore the integration of the fire duration variable, excluded in this study to preserve model impartiality, as a secondary prediction target once ignition risk has been established, thereby extending the framework from a binary danger classifier to a more comprehensive operational decision support tool. Finally, given the consistently strong performance of GENCLASS, further research should focus on refining and extending the interpretability of the generated rule sets, potentially enabling their direct adoption by fire prevention services as a practical, region specific alternative to the traditional 30/30/30 rule.

Author Contributions: Conceptualization, C.K..; methodology, C.K.; software, I.G.T.; validation, V.C.; formal analysis, V.C.; resources, G.K.; visualization, C.K.; supervision, I.G.T. All authors have read and agreed to the published version of the manuscript.

Funding: This research has been financed by the European Union : Next Generation EU through the Program Greece 2.0 National Recovery and Resilience Plan , under the call RESEARCH – CREATE – INNOVATE, project name "iCREW: Intelligent small craft simulator for advanced crew training using Virtual Reality techniques" (project code:TAEDK-06195).

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The original contributions presented in this study are included in the article. Further inquiries can be directed to the corresponding author.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. AA.com.tr. Southern California wildfires fully contained after 24 days (2025). Available from: https://www.aa.com.tr/en/americas/southern-california-wildfires-fully-contained-after-24-days/3468561 (Last accessed on 01/05/2026).
2. Bouvier, V., Dedieu, K., Lyons, D., & Raynaud, R. (2025). When it burns in winter: Modelling off-season California wildfires. Available from: https://hal.science/hal-05240719/(Last accessed on 01/05/2026).
3. Visual crossing.com. Available from: https://www.visualcrossing.com/weather-query-builder/ (Last accessed on 01/05/2026).
4. NASA. Wildfires Char South Korea. Available from: https://science.nasa.gov/earth/earth-observatory/wildfires-char-south-korea-149551/(Last accessed on 01/05/2026).
5. WeatherSpark. Available from: https://weatherspark.com/h/d/142033/2022/3/4/Historical-Weather-on-Friday-March-4-2022-in-Seoul-South-Korea (Last accessed on 01/05/2026).
6. Modis NASA. May 25, 2025 - Fires Continue to Scorch Russia's Far East. Available from: https://modis.gsfc.nasa.gov/gallery/individual.php?db_date=2025-05-25 (Last accessed on 01/05/2026).
7. WeatherSpark. Available from: https://weatherspark.com/d/127538/5/22/Average-Weather-on-May-22-in-Chita-Russia(Last accessed on 01/05/2026).
8. Suzuki, S., Manzello, S.L. The 2025 Wildland-Urban Interface (WUI) Fire Disaster in Ofunato City, Japan. Fire Technol 62, 49 (2026).
9. WeatherSpark. Available from: https://weatherspark.com/d/144114/2/26/Average-Weather-on-February-26-in-%C5%8Cfunato-Japan (Last accessed on 01/05/2026).
10. Diaz Cuevas, M. D. P., & Limones Rodriguez, N. (2021). El papel de la regla 30/30/30 en los incendios forestales. El caso de las provincias de Huelva y Sevilla. Available from: https://idus.us.es/server/api/core/bitstreams/7c154de8-c760-442d-b95b-9b9d5a79944d/content(Last accessed on 12/05/2026).
11. Cortez, P., & Morais, A. D. J. R. (2007). A data mining approach to predict forest fires using meteorological data. New Trends in Artificial Intelligence, Proceedings of the 13th EPIA, 512-523.
12. Yunhao, C., Jing, L., & Guangxiong, P. (2007). Forest fire risk assessment combining remote sensing and meteorological information. New Zealand Journal of Agricultural Research, 50(5), 1037-1044.
13. Sakr, G. E., Elhajj, I. H., & Mitri, G. (2011). Efficient forest fire occurrence prediction for developing countries using two weather parameters. Engineering Applications of Artificial Intelligence, 24(5), 888-894.
14. Agarwal, P., Tang, J., Narayanan, A. N. L., & Zhuang, J. (2020). Big data and predictive analytics in fire risk using weather data. Risk analysis, 40(7), 1438-1449.
15. Masinda, M. M., Li, F., Qi, L., Sun, L., & Hu, T. (2022). Forest fire risk estimation in a typical temperate forest in Northeastern China using the Canadian forest fire weather index: case study in autumn 2019 and 2020. Natural hazards, 111(1), 1085-1101.
16. Karali, A., Varotsos, K. V., Giannakopoulos, C., Nastos, P. P., & Hatzaki, M. (2023). Seasonal fire danger forecasts for supporting fire prevention management in an eastern Mediterranean environment: the case of Attica, Greece. Natural hazards and earth system sciences, 23(2), 429-445.
17. Parvar, Z., Saeidi, S., & Mirkarimi, S. (2024). Integrating meteorological and geospatial data for forest fire risk assessment. Journal of environmental management, 358, 120925.
18. Global Forest Watch. Available from: https://www.globalforestwatch.org/dashboards/country/GRC/3/?category=fires&location=WyJjb3VudHJ5IiwiR1JDIiwiMyJd&map=eyJjYW5Cb3VuZCI6dHJ1ZX0%3D (Last accessed on 02/05/2026).
19. Tsoulos, I. G., Charilogis, V., Kyrou, G., Stavrou, V. N., & Tzallas, A. (2025). OPTIMUS: A Multidimensional Global Optimization Package. Journal of Open Source Software, 10(108), 7584.
20. Tsoulos, I. G. (2022). QFC: A Parallel Software Tool for Feature Construction, Based on Grammatical Evolution. Algorithms, 15(8), 295.
21. Anastasopoulos, N., Tsoulos, I. G., & Tzallas, A. (2021). GenClass: A parallel tool for data classification based on Grammatical Evolution. SoftwareX, 16, 100830.
22. Kopitsa, C., Tsoulos, I. G., Charilogis, V., & Stavrakoudis, A. (2024). Predicting the duration of forest fires using machine learning methods. Future Internet, 16(11), 396.
23. Müller, B., Reinhardt, J., & Strickland, M. T. (2012). Neural networks: an introduction. Springer Science & Business Media.
24. Gurney, K. (2018). An introduction to neural networks. CRC press.

25. M.J.D Powell, A Tolerant Algorithm for Linearly Constrained Optimization Calculations, Mathematical Programming **45**, pp. 547-566, 1989.
26. J. Park and I. W. Sandberg, Universal Approximation Using Radial-Basis-Function Networks, Neural Computation **3**, pp. 246-257, 1991.
27. G.A. Montazer, D. Giveki, M. Karami, H. Rastegar, Radial basis function neural networks: A review. Comput. Rev. J **1**, pp. 52-74, 2018.
28. S.J. Rigatti, Random forest. Journal of Insurance Medicine **47**, pp. 31-39, 2017.
29. Dimitris Gavrilis, Ioannis G. Tsoulos, Evangelos Dermatas, Selecting and constructing features using grammatical evolution, Pattern Recognition Letters **29**,pp. 1358-1365, 2008.

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.